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Motor and Gearbox Matching Checklist

For OEM designers and maintenance engineers specifying or troubleshooting geared DC drives

Complete each step with measured or datasheet values rather than assumptions. Any item that fails points to the component or setting that must change before the drive can deliver its required output.

Step 1 — Define the output requirement

- [ ] Running torque at the output shaft in Nm, at the worst-case operating condition.
- [ ] Breakaway torque from rest, measured with a torque wrench or spring balance where possible.
- [ ] Required output speed and its tolerance, for example 50 rpm ± 5 %.
- [ ] Duty: S1, S2 or S3, hours per day, starts per hour and any reversing.
- [ ] Load character and service factor: uniform 1.0–1.25, moderate shock 1.25–1.5, heavy shock 1.75–2.0.
- [ ] Load inertia referred to the output shaft in kg·m², and the required acceleration time.

Step 2 — Verify gearbox efficiency

- [ ] Obtain efficiency at your load, speed and temperature, not only the best catalogue figure.
- [ ] Spur or helical: about 95–98 % per well-made stage; expect less in small greased gearheads and at light load.
- [ ] Worm: obtain both running and starting efficiency for the actual ratio and number of starts.
- [ ] Use starting efficiency for breakaway checks and running efficiency for continuous checks.
- [ ] Allow for cold lubricant if the machine starts in low ambient temperatures.

Step 3 — Check the motor operating point

- [ ] Motor torque = output torque / (ratio × efficiency).
- [ ] Continuous motor torque should not exceed about 70–90 % of rated torque.
- [ ] Loaded motor speed = no-load speed × (1 − motor torque / stall torque).
- [ ] Output speed = loaded motor speed / ratio; confirm it is within tolerance.
- [ ] Motor current = no-load current + motor torque / Kt; compare with rated current.
- [ ] Breakaway motor current must be below the controller current limit.

Step 4 — Protect the gearbox

- [ ] Gearbox continuous rating ≥ load torque × service factor.
- [ ] Compare motor stall torque × ratio × efficiency with the gearbox peak rating; if higher, a current limit or slip clutch is mandatory.
- [ ] Maximum current limit = no-load current + gearbox peak torque / (Kt × ratio × efficiency).
- [ ] Motor-side brake torque × ratio × efficiency must not exceed the gearbox peak rating.
- [ ] Check stopping and end-of-travel shocks: rotor inertia appears multiplied by the ratio squared at the output.

Step 5 — Thermal and shaft-load checks

- [ ] Gearbox heat = input power × (1 − efficiency); compare with the maker's thermal rating.
- [ ] Housing temperature rise = gearbox heat × housing thermal resistance; keep within lubricant and seal limits.

- [ ] Overhung load = 2 × output torque / sprocket or pulley pitch diameter × transmission factor.
- [ ] Mount sprockets and pulleys as close to the housing as possible and confirm the permitted load position.
- [ ] Check axial thrust from helical or worm gears and from the driven machine.

Step 6 — Commissioning measurements

- [ ] Record terminal voltage and motor current at running load and at breakaway.
- [ ] Record output speed under load with a tachometer and compare with the calculation.
- [ ] Record motor and gearbox housing temperatures after thermal stabilisation.
- [ ] Repeat the breakaway test after an overnight cold soak.
- [ ] File all readings as the baseline for maintenance trending.

Gear type sizing reference (typical planning values)

Gear type	Running efficiency	Starting efficiency	Main sizing risk
Spur or helical, per stage	about 95–98 %	slightly lower	Light-load drag in small units
Planetary, per stage	about 95–97 %	slightly lower	Peak torque on small planets
Worm, multi-start	about 70–90 %	noticeably lower	Thermal capacity
Worm, single-start high ratio	about 40–60 %	can fall to 30–45 %	Breakaway torque and heat

Efficiency figures are typical ranges for planning only. Always use the gearbox manufacturer's data for the specific ratio, size, lubricant and operating temperature.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.